

The Hierarchical Observer and the Hyper-Prior

A Bayesian reformulation of the Switching observer (Hypothesis 1)

Neuromatch Bayesian Modelling project

The one idea: priors on priors

A *hierarchical* model just means the prior has parameters, and those parameters themselves have a prior — the **hyper-prior**. In a flat Bayesian model you write $\text{posterior} \propto \text{likelihood} \times \text{prior}$ with the prior fully known. In a hierarchical model you admit you do not know the prior exactly: you treat its parameters as latent variables and place a distribution over them. A hyper-prior is literally *a prior over the parameters of a prior*.

The figure below is the generative story for this task. Reading it top-down: fixed hyperparameters η set the prior’s parameters — the mixing weight π , the mode μ , and the concentration k ; each trial draws a latent cause z , then a direction θ , then a sensory measurement m , which the observer reads out as an estimate $\hat{\theta}$.

This document builds the model up one level at a time and shows, mathematically, how the paper’s Switching observer falls out as a special case.

Level 1: the latent cause turns the prior into a mixture

This is the part Laquittaine & Gardner (2018) propose in their Discussion — that “motion directions were drawn from either a uniform or a peaked distribution at each trial.” Introduce a per-trial latent indicator $z_t \in \{\text{peaked}, \text{uniform}\}$ with $P(z = \text{peaked}) = \pi$. The direction is drawn from whichever component z selects:

$$\theta \mid z = \text{peaked} \sim \text{VM}(\mu, k), \quad \theta \mid z = \text{uniform} \sim \mathcal{U}(0, 360).$$

Marginalising out z gives the **effective prior** — a von Mises sitting on a uniform floor:

$$p(\theta) = \pi \text{VM}(\theta; \mu, k) + (1 - \pi) \frac{1}{360}.$$

That is the whole reformulation: the “switch” becomes a latent variable in a generative model rather than an ad-hoc rule. (Note this is the same object as the paper’s

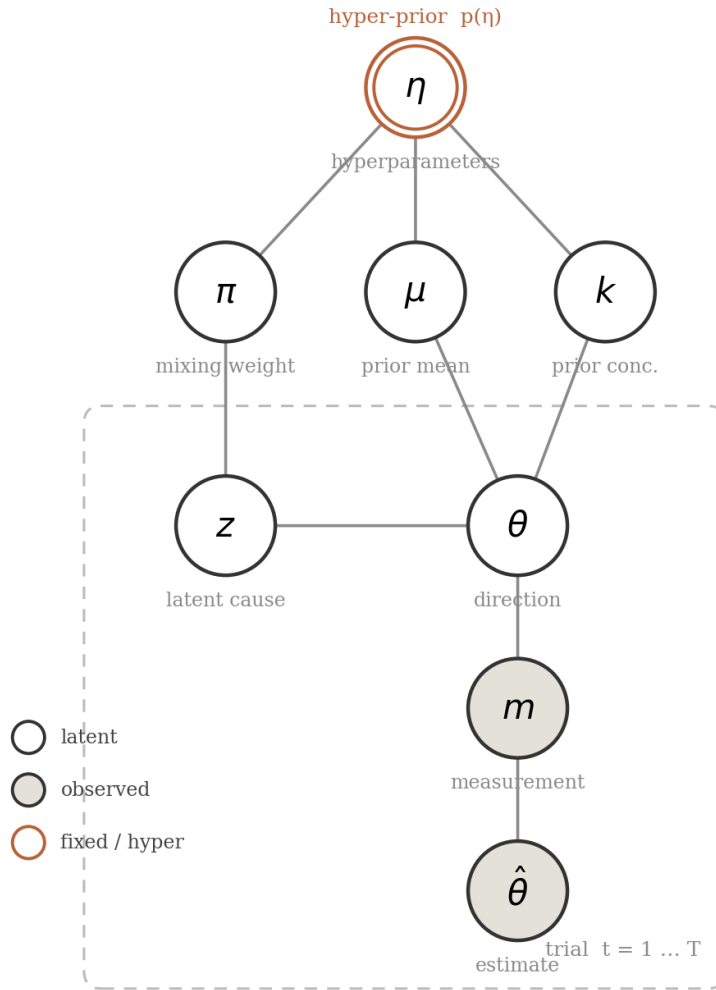


Figure 1: Hierarchical generative model. Circles are variables; shaded = observed, brown double ring = fixed hyperparameters governed by a hyper-prior. The plate repeats the per-trial variables over trials.

long-tailed prior, but here it arises from a latent-cause hierarchy rather than being posited directly.)

Inference: the posterior is also a mixture, weighted by evidence

Because the prior is a mixture, the posterior over θ given a measurement m splits into two **sub-posteriors**, one per cause:

$$p(\theta | m) = r(m) \underbrace{p(\theta | m, \text{peaked})}_{\propto p(m|\theta) \text{VM}(\theta; \mu, k)} + (1 - r(m)) \underbrace{p(\theta | m, \text{uniform})}_{\propto p(m|\theta)}.$$

The peaked sub-posterior is the ordinary Bayesian-integration posterior (likelihood $\times$ von Mises prior — the Girshick posterior in the code); the uniform sub-posterior is just the likelihood, because a uniform prior contributes nothing. The mixing coefficient $r(m)$ is the **responsibility** — the posterior probability that this trial came from the peaked cause — and it is itself Bayes’ rule one level up:

$$r(m) = P(z = \text{peaked} | m) = \frac{\pi p(m | \text{peaked})}{\pi p(m | \text{peaked}) + (1 - \pi) p(m | \text{uniform})}.$$

The quantities $p(m | z) = \int p(m | \theta) p(\theta | z) d\theta$ are the **model evidences** (marginal likelihoods) of the two hypotheses. For von Mises components,

$$p(m | \text{peaked}) = \int \text{VM}(m; \theta, k_{llh}) \text{VM}(\theta; \mu, k) d\theta \approx \text{VM}(m; \mu, \tilde{\kappa}), \quad p(m | \text{uniform}) = \frac{1}{360}.$$

So the responsibility is high when the measurement lands near the prior mode (225°) or the peaked component is sharp, and low when the measurement is far from 225° or the prior is weak. **This is a graded, stimulus-dependent switch** — precisely what the fixed Switching observer cannot express.

The Switching observer is a special case

The paper’s Switching observer freezes the mixing weight to a concentration ratio,

$$p_{\text{prior}} = \frac{k_{\text{prior}}}{k_{\text{prior}} + k_{llh}}.$$

That is exactly $r(m)$ with the measurement’s *location* discarded: you compare the

two evidences using only how concentrated each component is, not where m fell. So the Switching observer is the “flat,” measurement-blind version of the hierarchical responsibility. Computing $r(m)$ per trial (Hypothesis 1) is strictly richer — and it is why Deck B’s finding that prior reliance depends *non-monotonically* on prior width breaks the frozen ratio but is perfectly consistent with the full $r(m)$.

Where the bimodality comes from: the readout of z

Given the mixture posterior you can collapse it to an estimate in three ways, and they behave very differently:

Model averaging — take the mean/mode of the whole mixture. Smooth, essentially the Basic Bayesian observer; this *loses* the bimodality.

Model selection — report the mode of whichever sub-posterior has higher $r(m)$. Winner-take-all.

Probability matching / sampling — draw $z \sim \text{Bernoulli}(r(m))$ and report the chosen sub-posterior. Across trials this places mass at 225° (when the peaked cause wins) *and* at the likelihood peak (when the uniform cause wins) — the **bimodal estimate distribution** that motivates the whole project.

So bimodality = a mixture posterior + a non-averaging readout of the latent cause. The Switching observer is essentially selection/matching.

Level 2: the hyper-prior, and why you need it

Everything above assumed the prior’s parameters (π, μ, k) were known. But subjects must *learn* them. The hyper-prior $p(\eta)$ over $\eta = (\pi, \mu, k)$ makes them latent variables you infer from the trial stream — this is hierarchical (empirical) Bayes. Formally, evidence about the hyperparameters accumulates as

$$p(\eta \mid m_{1:t}) \propto p(\eta) \prod_{s \leq t} p(m_s \mid \eta),$$

and the prior the observer actually uses on the next trial is the **predictive prior**, already shaped by experience:

$$p(\theta_{t+1} \mid m_{1:t}) = \int p(\theta_{t+1} \mid \eta) p(\eta \mid m_{1:t}) d\eta.$$

Three consequences connect directly to the project’s other models. First, as trials accumulate the posterior over k sharpens, so the learned prior gets stronger over time — exactly the paper’s Prior Learning Model $k_{prior} (1 - e^{-t/\tau})$ and Deck A’s on-line update $\mu \leftarrow \alpha(\theta_{true} - \mu)$; both are cheap point-estimate approximations to

this integral (online learning is a one-sample stochastic approximation of hierarchical inference). Second, if η is allowed to *change* over time (a new block, a volatile world), the hyper-prior carries a change process and the exact online inference becomes Bayesian online changepoint detection — the principled version of Hypothesis 2. Third, in Lee & Mumford’s (2003) cortical reading, each visual area’s estimate is the prior (the contextual hyper-prior) for the area below it, with feedback carrying priors downward and feedforward carrying evidence upward — a physical hierarchy of priors-on-priors implemented by belief propagation.

What changes in the code

In `girshick_lookup` (notebook §2; the “Bayesian integration” step in `Model_explainer`), replace the single von Mises prior with the mixture $\pi \text{VM} + (1 - \pi)/360$, additionally compute $r(m)$ from the two evidences, and pick a readout (average / select / sample). Learning the hyperparameters is the notebook’s §8 online loop, or a filter/changepoint model over (μ, k, π) . Setting $r(m) = k/(k + k_{lh})$ and freezing (μ, k) collapses the whole construction back to the Switching observer — your baseline.

References

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